SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I G. Yeshvanth Ram, 192311397 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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Col Assessment tool 03

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1 Smart Home Automation System - Intelligent Agents:

Introduction:

A smart home automation system controls lighting, temperature, security using Sensors such as motion detectors, temperature sensors, and time of day information. Intelligent agents help the system make decision automatically to improve Comfort energy efficiency and security.

Types of Intelligent Agents:

(a) Simple Reflex Agent:

- Acts only on the current sensor input using predefined rules
- Does not remember previous states.

Example:

- if motion is detected → Turn ON Lights
- if temperature > 30°C → Turn ON air conditioner
- if no motion for 10 minutes → Turn OFF Lights.

Advantages:

- Fast Response
- Easy to implement
- Suitable for simple task

Disadvantages

- Cannot learn user habits
- Cannot make decisions using past information

(b) Model-Based Agent:

- Maintains an internal model of the environment.
- Uses previous information along with current sensor data.

Example:

- knows that the bedroom is occupied even if motion is temporarily absent
- Detects an open door at night and raises a security alert.

Advantages:

- Handles partially observable environment
- More accurate than a simpler reflex agent.
- Improves Security and reliability

Disadvantages:

- Requires additional memory
- More complex to design

c) Goal based Agent:

- Makes decision to achieve predefined goals.

Goals:

* Maintain room temperature between 22-25°C
* Reduce electricity consumption.
* Keep the home secure.

Example:

- Choose whether to use a fan or AC to achieve the desired temperature with minimum energy consumption.

Advantages:

- flexible decision making.
- Can Evaluate different Action
- Suitable for automation with multiple objectives

Disadvantages:

- Requires Planning
- Slower than reflex Agent.

d) Utility Based Agent:

- Choose the action that provides the highest overall benefit (Utility)
- Balance multiple factors such as comfort, cost and security.

Example:

- Uses Natural ventilation instead of AC when weather is pleasant
- lower heating when electricity prices are high
- Gives higher priority to security alarms than lighting.

Advantages:

- Optimizes comfort and energy efficiency
- Handles conflicting objectives
- Produces the best overall decisions

Disadvantages:

- Complex Utility Calculation.
- Higher computational cost.

Best Agent for Smart Home.

A hybrid Agent (Model-Based + Utility-Based + Learning Agent) is the most effective.

2. Robot in an Unknown Maze - UCS and IDS

Problem Statement:

A robot must find the exit of an unknown maze without knowing the layout in advance

Uniform Cost Search (UCS)

Working:

- Expands the node with the lowest cumulative path cost.
- Guarantees the least cost path

Advantages:

- Complete.
- Optimal when step costs differ
- finds the shortest-cost solution.

Disadvantages.

- Requires large memory
- Can be slow for large search spaces

Complexity

- Time: Exponential (Approximately $O(b^{1+C^*/\epsilon})$
- Space: Exponential.

Iterative Deepening Search (IDS)

Working:

- Repeatedly performs Depth-Limited Search with increasing depth limits
- Combines DFS's low memory with BFS's completeness.

Advantages:

- Complete
- Very low memory usage
- Suitable when solution depth is unknown.

Disadvantage:

- Re-expands nodes multiple times
- Slower than BFS when the solution is deep.

Complexity

- Time = $O(b^d)$, Space = $O(bd)$

Intelligent Agent Relation.

Simple Reflex Agent

- Only reacts to nearby obstacles.
- Cannot plan a complete route

Model Based Agent:

- Builds a map while exploring
- Avoids revisiting explored paths.

Goal Based Agent:

- Plans actions to reach the maze exits

Utility Based Agent:

- Choose routes that minimize time, distance and energy.

Best combination:

Model Based Goal-Based Agent + Uniform cost search (UCS)

- Builds knowledge of the maze
- Plans efficiently toward the exits
- find the Least-cost path.
- Avoids unnecessary exploration.

Smart traffic Management System - Intelligent Agents:

Introduction:

A smart traffic system management system Controls traffics signals using vehicle density pedestrian movement, wheater Condition and historical traffic data to reduces congestion and travell time.

Types of Intelligent Agents

a) Simple Reflex Agent:

Acts only on current sensor inputs

Example:

- If vehicle count is high → Extend green signal.
- If pedestrian button is pressed → Activate pedestrian crossing.

Advantage:

- Fast response
- Easy implementation

Disadvantages:

- Cannot predict future traffic
- No learning Capability.

(b) Model Based Agent:

Maintains information about traffic conditions.

Example:

- Tracks congestion at nearby intersection.
- Predict queue buildup based on previous observation.

Advantage:

- Better coordination between Signals.
- More accurate decisions.

Disadvantages:

- Requires additional storage and Computation.

c) Goal - Based Agent.

- Works towards specific goal.

Goals:

- Reduce waiting time
- Minimize Congestion.
- Ensure pedestrian safety.

Example:

· Adjust signal timing to clear heavy traffic first.

Advantage:

- Flexible planning.
- Supports multiple objectives.

Disadvantages.

- More computationally intensive

d, Utility - Based Agents.

Choose action with the highest overall benefits.

Utility factors

- Vehicle waiting time
- Fuel consumption.
- Emergency vehicle priority
- Pedestrian safety.
- Weather conditions.

Example:

- Gives priority to ambulance
- Extends green signals during heavy rain to reduces Congestion.

Advantages:

- Balances multiple objectives
- Optimizes overall traffic flow

Disadvantages:

- Complex calculation.
- Require Significant processing power.

Best Agent for Smart traffic system:

A hybrid Agent (Model based + Utility - Based + learning Agent) is the most effective.